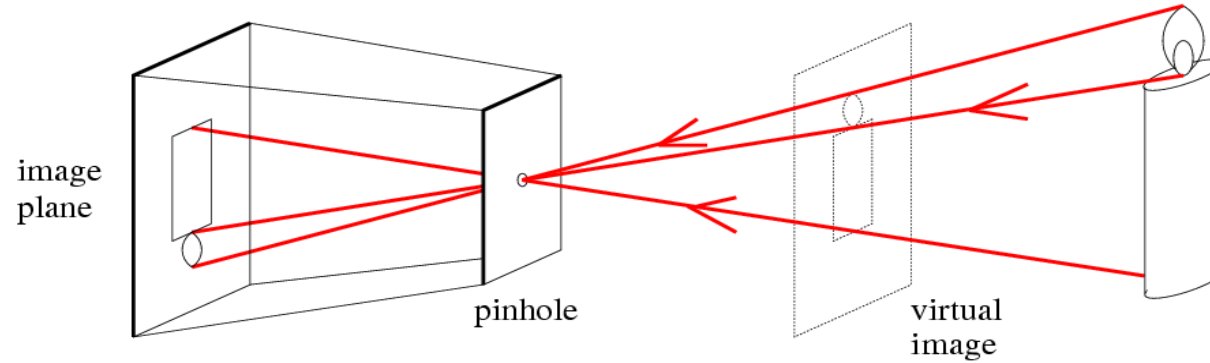


Lecture 1.2

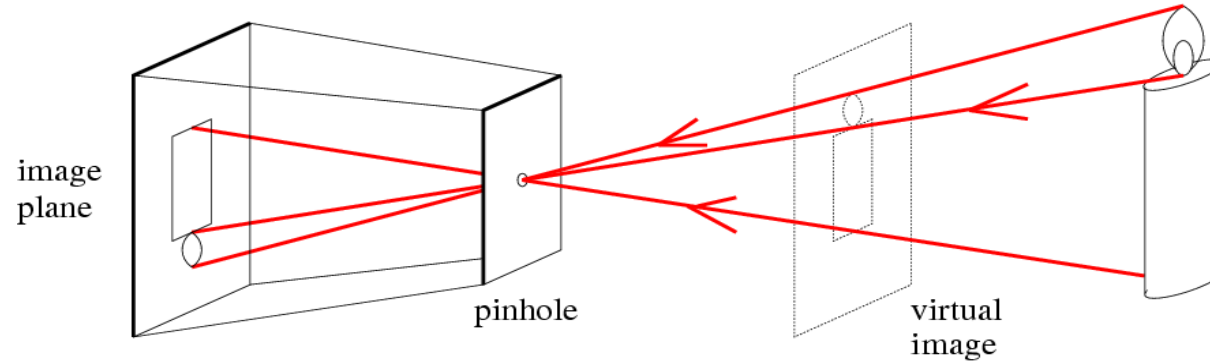
Pose in 2D and 3D

Motivation



- For the pinhole camera, the correspondence between observed 3D points in the world and 2D points in the captured image is given by straight lines through a common point (pinhole)
- This correspondence can be described by a mathematical model known as “*the perspective camera model*” or “*the pinhole camera model*”
- This model can be used to describe the imaging geometry of many modern cameras, hence it plays a central part in computer vision

Motivation



- Before we can study the perspective camera model in detail, we need to expand our mathematical toolbox
- We need to be able to mathematically describe the position and orientation of the camera relative to the world coordinate frame
- Also we need to get familiar with some basic elements of projective geometry, since this will make it MUCH easier to describe and work with the perspective camera model

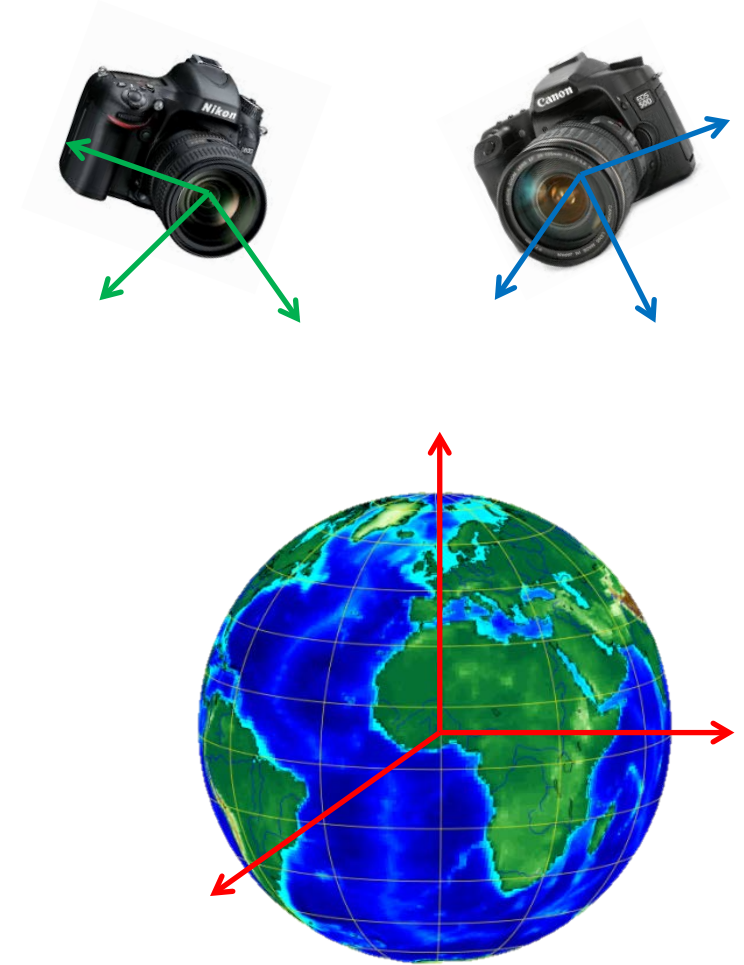
Introduction

- In computer vision we are often interested in the geometrical aspects of imaging
 - Points in the world $\leftrightarrow$ pixels in an image
 - Pixels in image 1 $\leftrightarrow$ pixels in image 2
- In order to express and study geometrical problems related to imaging, we first need to know how to describe the position and orientation of objects
- Position and orientation is together known as *pose*

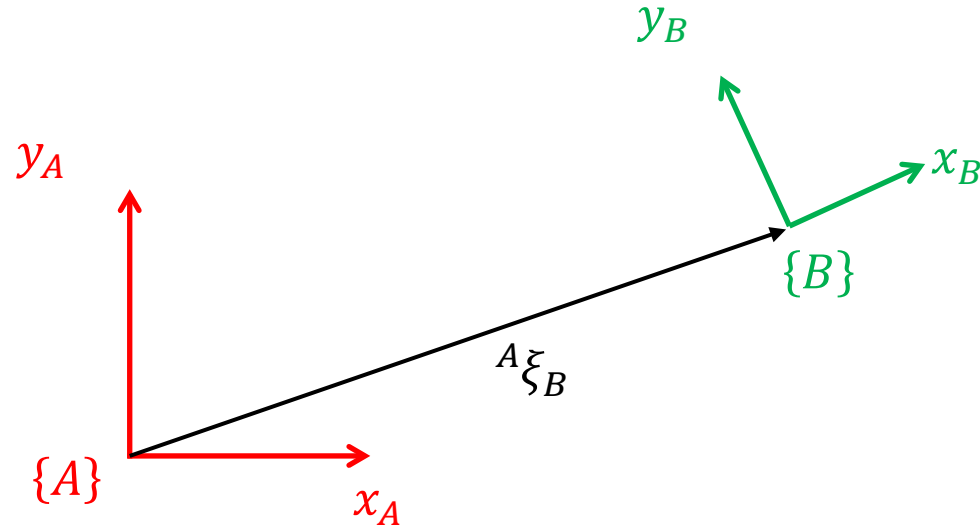


Introduction

- By representing all relevant objects by coordinate frames, it is possible to numerically represent the pose of one object relative to another
- In the following we will look at pose
 - General properties
 - Representation in 2D
 - Representation in 3D

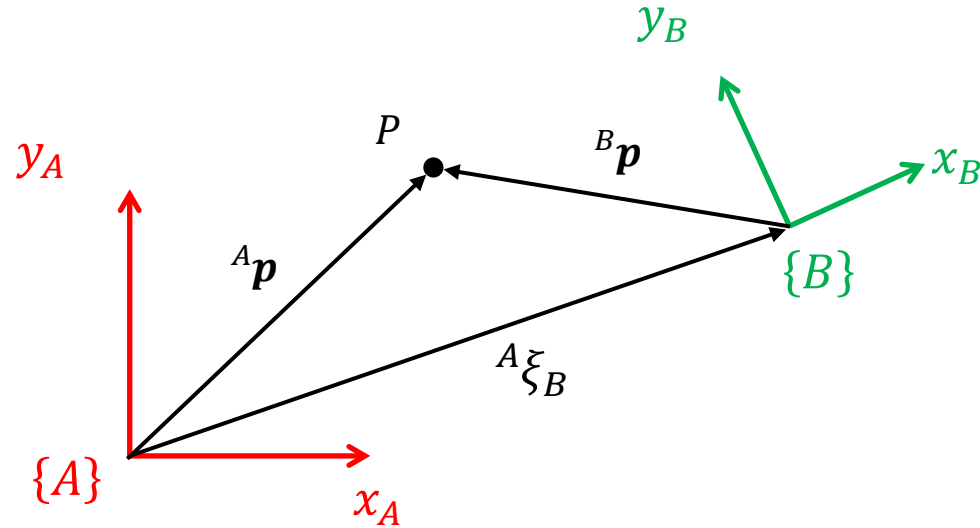


General properties



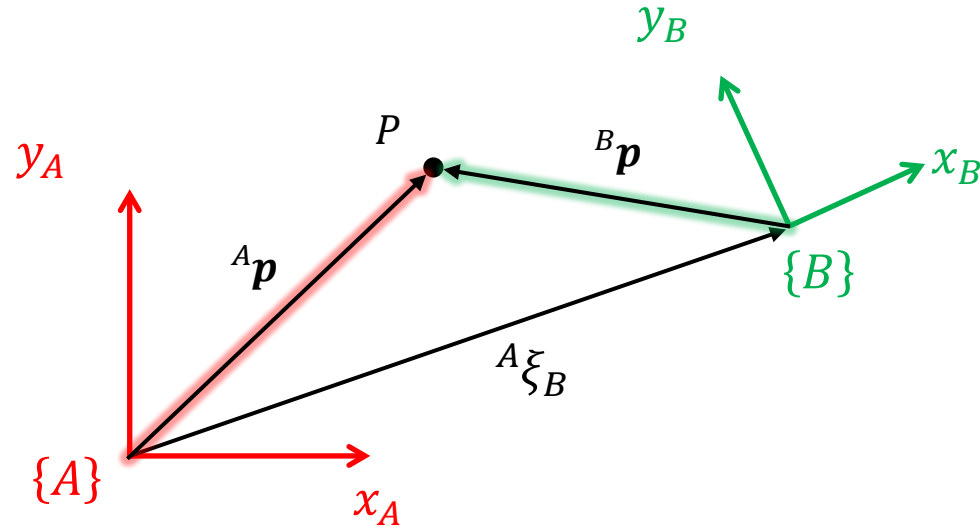
- Let us denote by ${}^A\xi_B$ the pose of frame $\{B\}$ relative to frame $\{A\}$
- We can think of ${}^A\xi_B$ as the translation and rotation required in order to make $\{A\}$ coincide with $\{B\}$

General properties



- A point P can be described with respect to either frame
- These descriptions are related by the pose
- Formally we write this as ${}^A\mathbf{p} = {}^A\boldsymbol{\xi}_B \cdot {}^B\mathbf{p}$

General properties



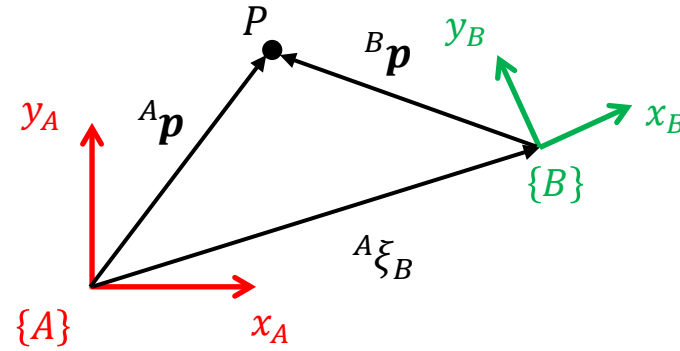
Note that ${}^A \mathbf{p}$ and ${}^B \mathbf{p}$ are different vectors!!!

- A point P can be described with respect to either frame
- These descriptions are related by the pose
- Formally we write this as ${}^A \mathbf{p} = {}^A \xi_B \cdot {}^B \mathbf{p}$

General properties

- Action on points

$${}^A\mathbf{p} = {}^A\boldsymbol{\zeta}_B \cdot {}^B\mathbf{p}$$



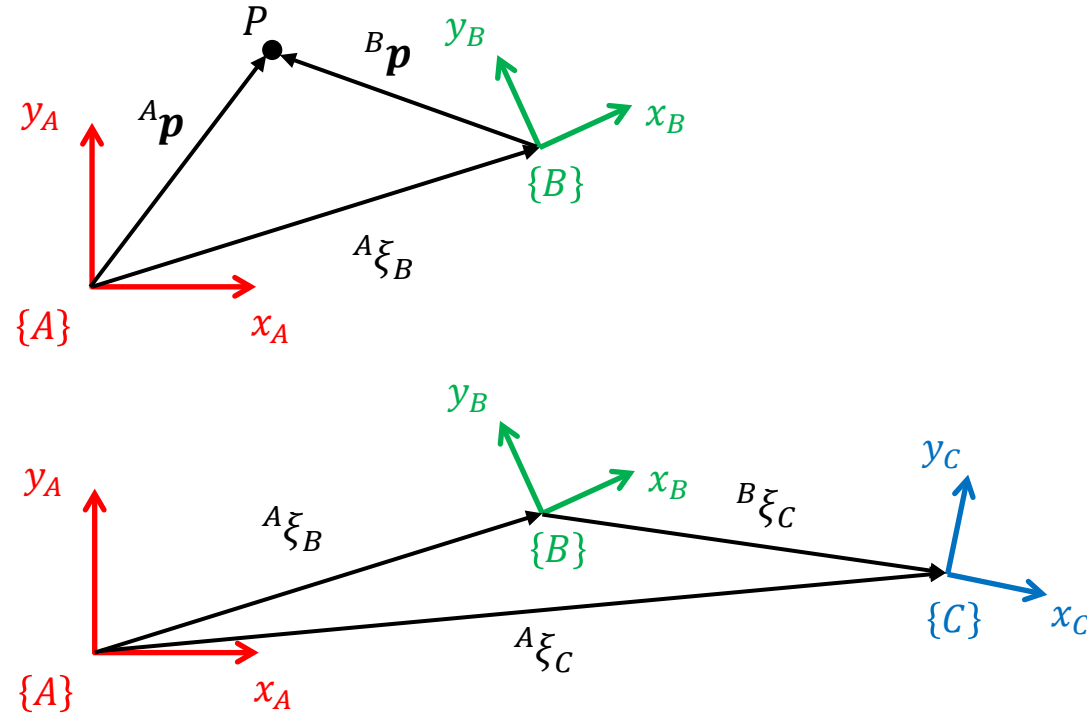
General properties

- Action on points

$${}^A\mathbf{p} = {}^A\xi_B \cdot {}^B\mathbf{p}$$

- Composition

$${}^A\xi_C = {}^A\xi_B \oplus {}^B\xi_C$$



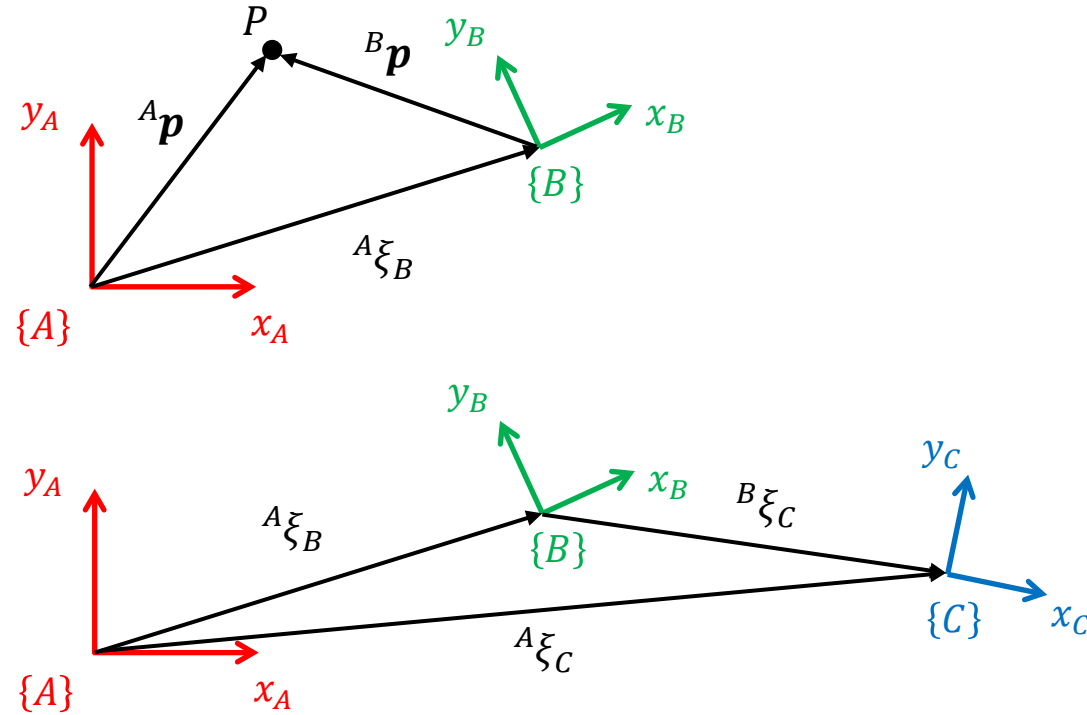
General properties

- Action on points

$${}^A\mathbf{p} = {}^A\xi_B \cdot {}^B\mathbf{p}$$

- Composition

$${}^A\xi_C = {}^A\xi_B \oplus {}^B\xi_C$$



What about ${}^A\xi_B \oplus {}^B\xi_A$?

General properties

- Action on points

$${}^A\mathbf{p} = {}^A\xi_B \cdot {}^B\mathbf{p}$$

- Composition

$${}^A\xi_C = {}^A\xi_B \oplus {}^B\xi_C$$

- Inverse

$$\ominus {}^A\xi_B = {}^B\xi_A$$

- Neutral element

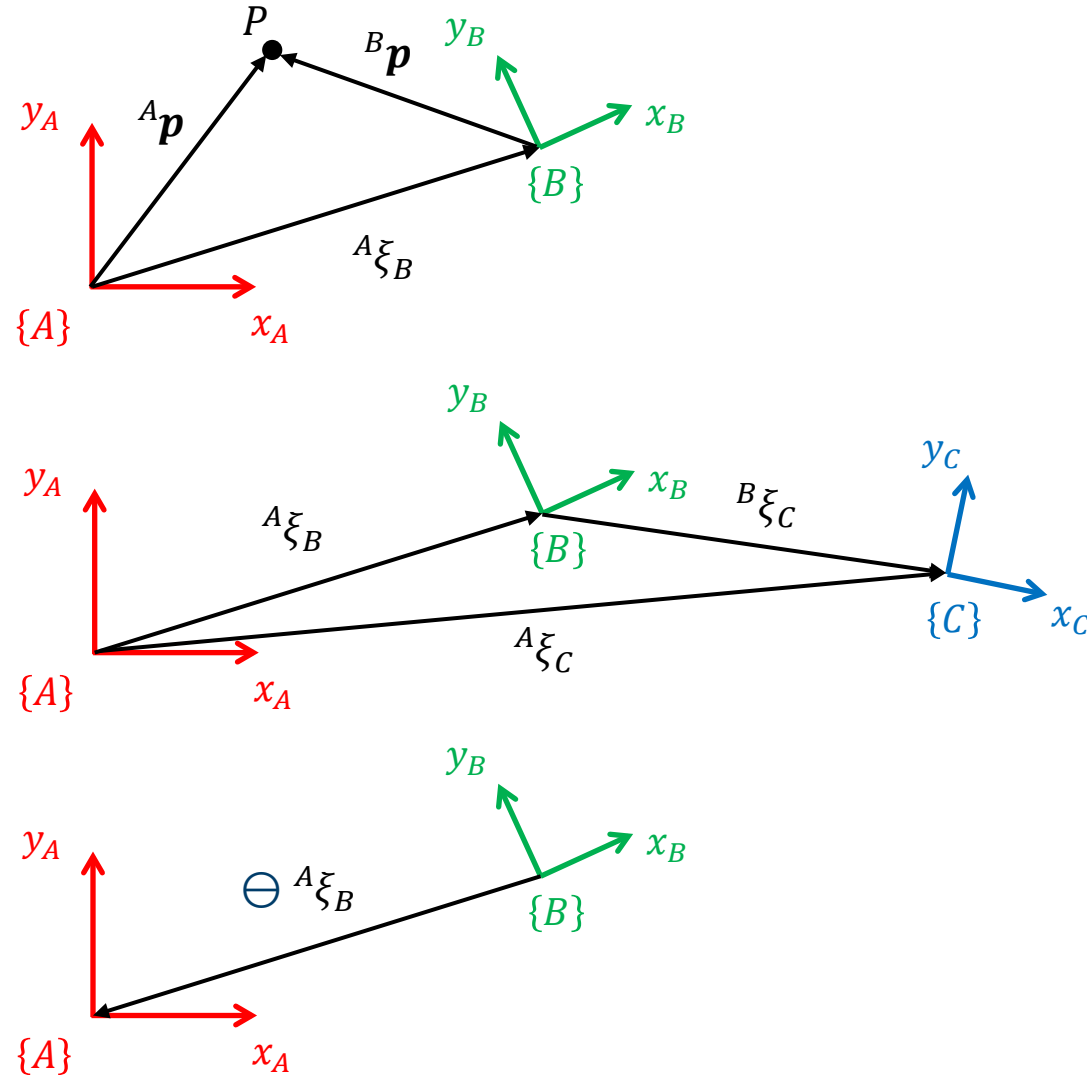
$$0 = {}^A\xi_A$$

$$0 \oplus \xi = \xi$$

$$\xi \ominus 0 = \xi$$

$$\xi \ominus \xi = 0$$

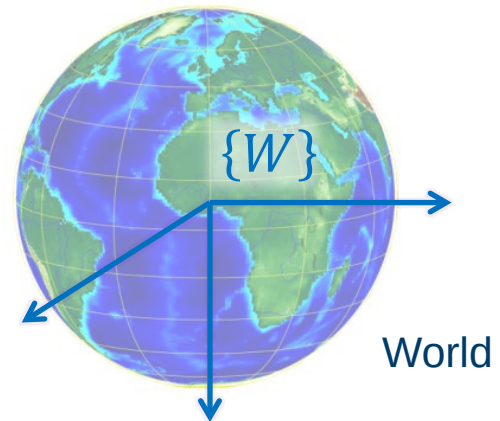
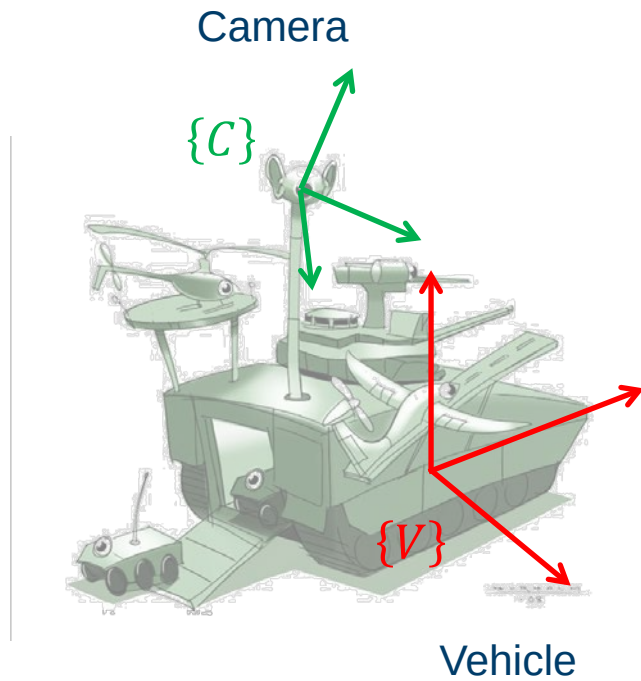
$$\ominus \xi \oplus \xi = 0$$



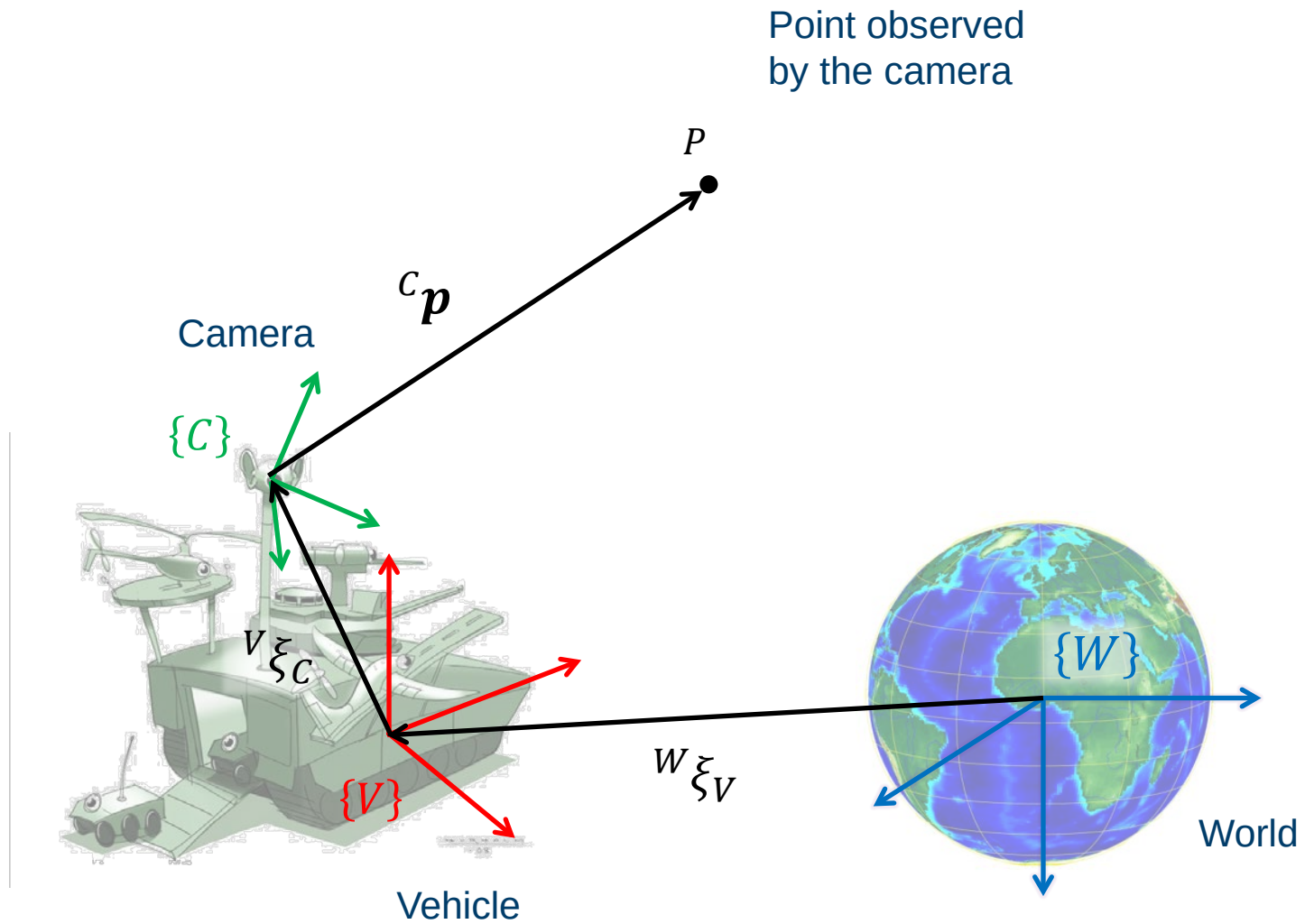
Example

Point observed
by the camera

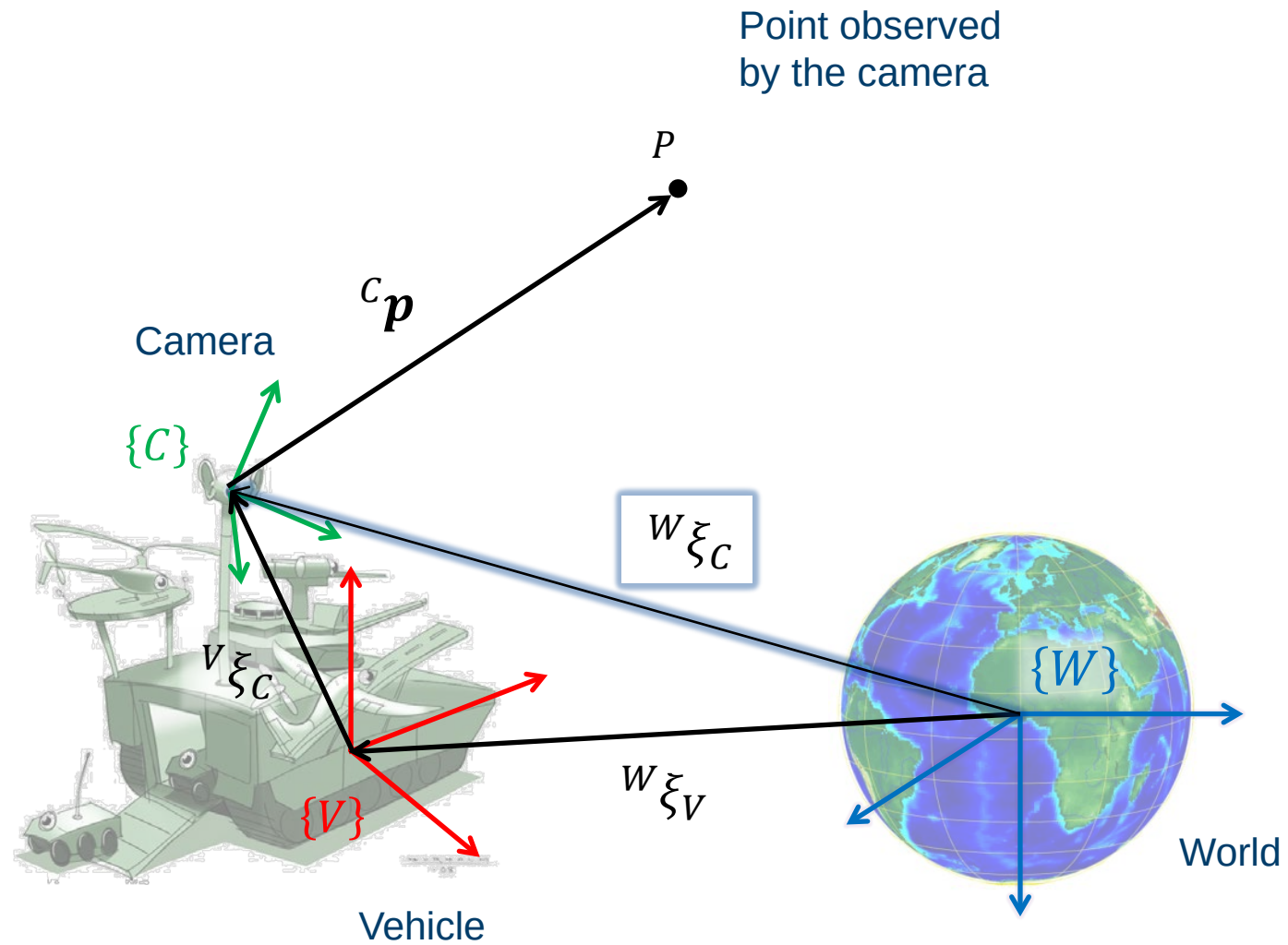
P



Example

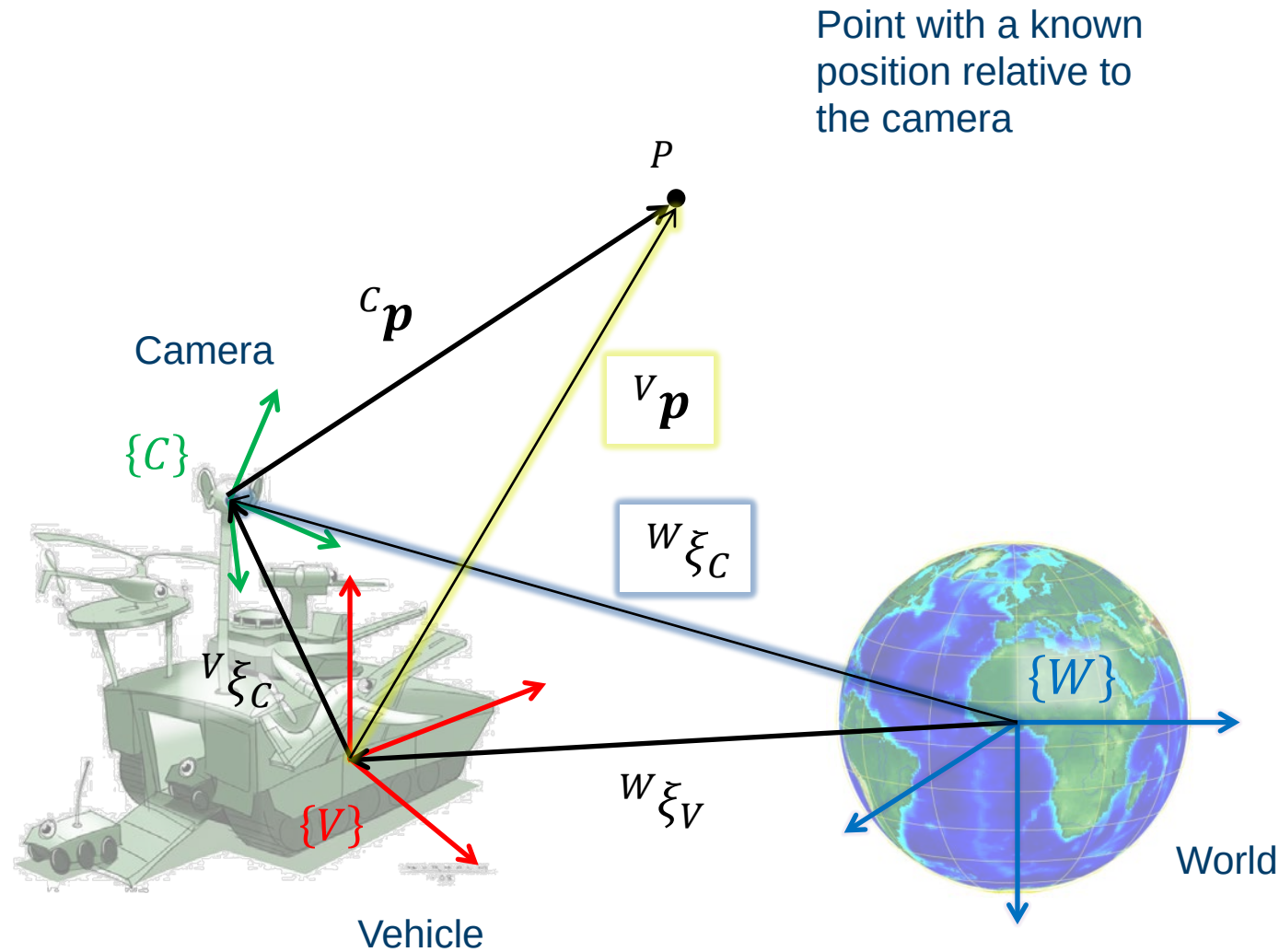


Example



$${}^W \xi_C = {}^W \xi_V \oplus {}^V \xi_C$$

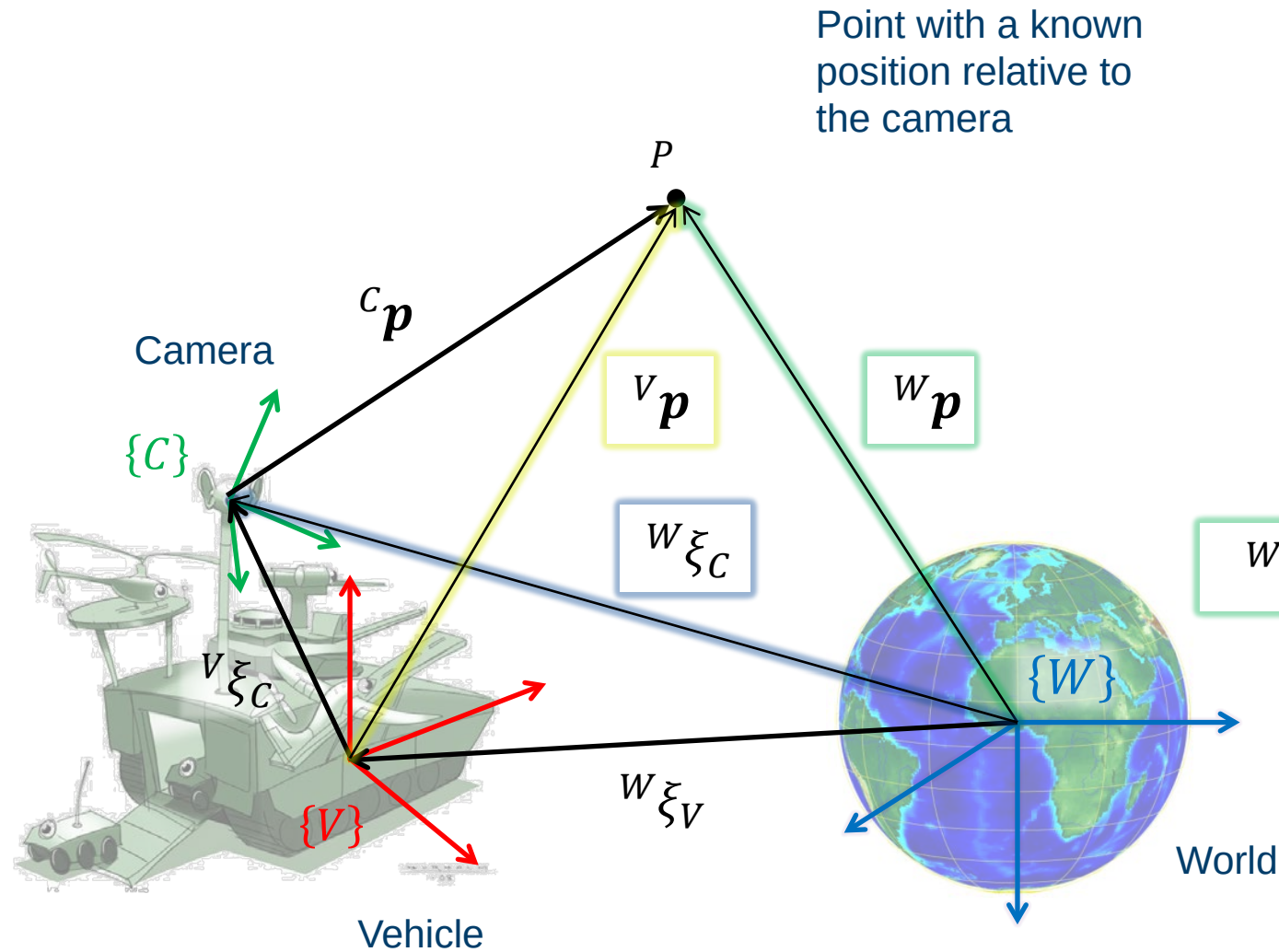
Example



$${}^W\xi_C = {}^W\xi_V \oplus {}^V\xi_C$$

$${}^Vp = {}^V\xi_C \cdot {}^Cp$$

Example

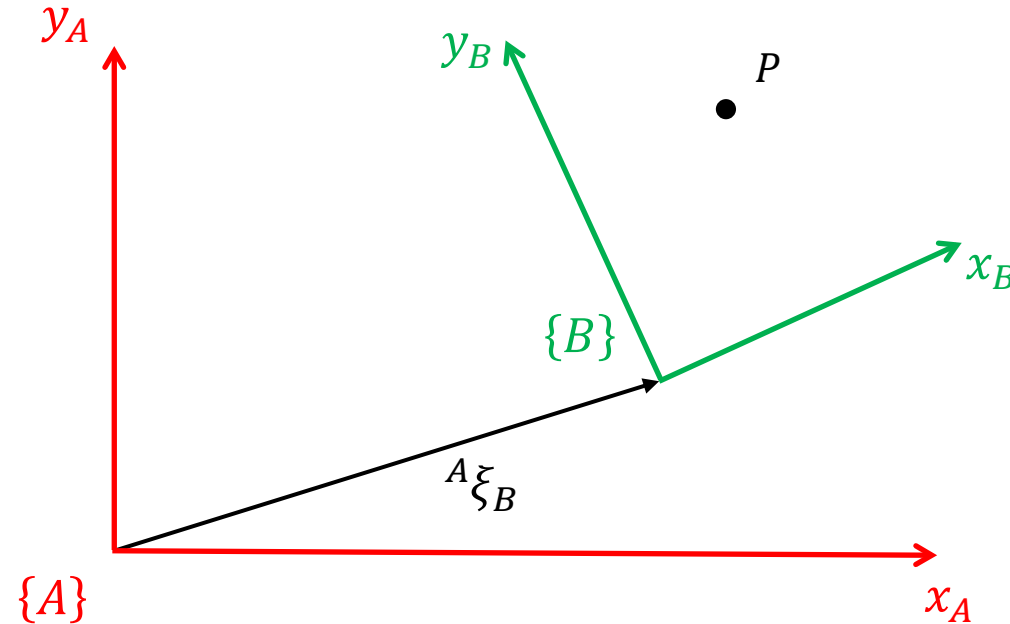


$$W_{\xi_C} = W_{\xi_V} \oplus V_{\xi_C}$$

$$v_p = V_{\xi_C} \cdot c_p$$

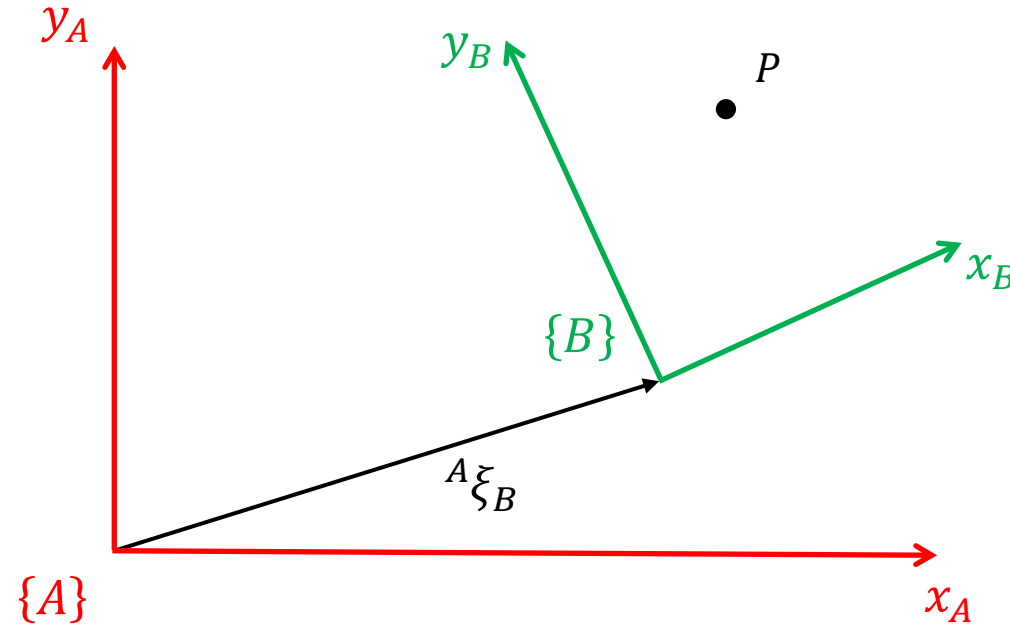
$$w_p = (W_{\xi_V} \oplus V_{\xi_C}) \cdot c_p$$

Investigating pose in 2D



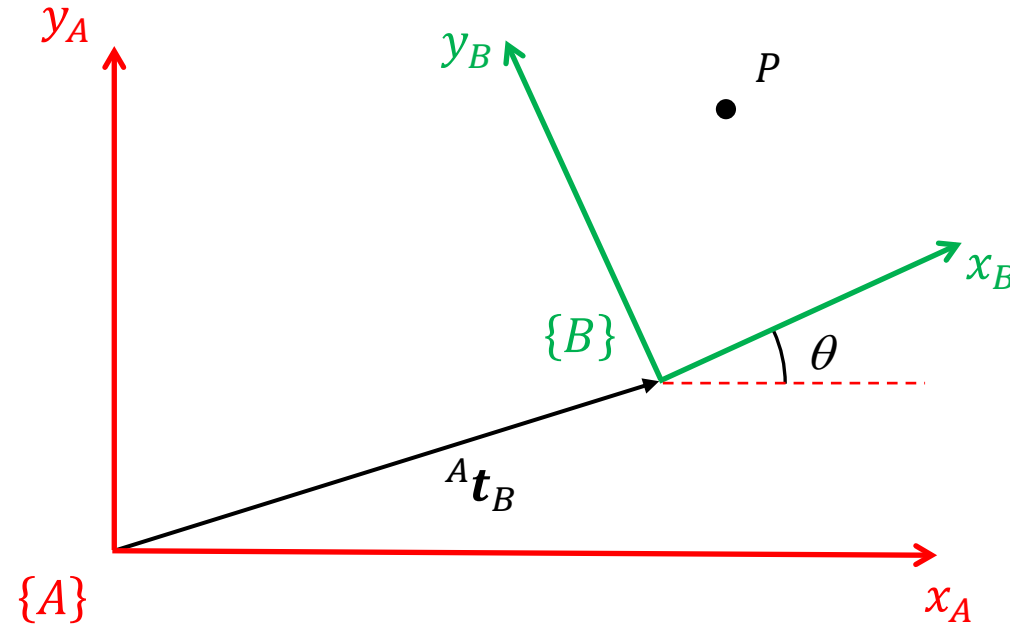
- Given two 2D frames, $\{A\}$ and $\{B\}$, how can we represent the pose ${}^A\xi_B$?
- We need a numerical representation in order to compute ${}^A\boldsymbol{p} = {}^A\xi_B \cdot {}^B\boldsymbol{p}$

Investigating pose in 2D



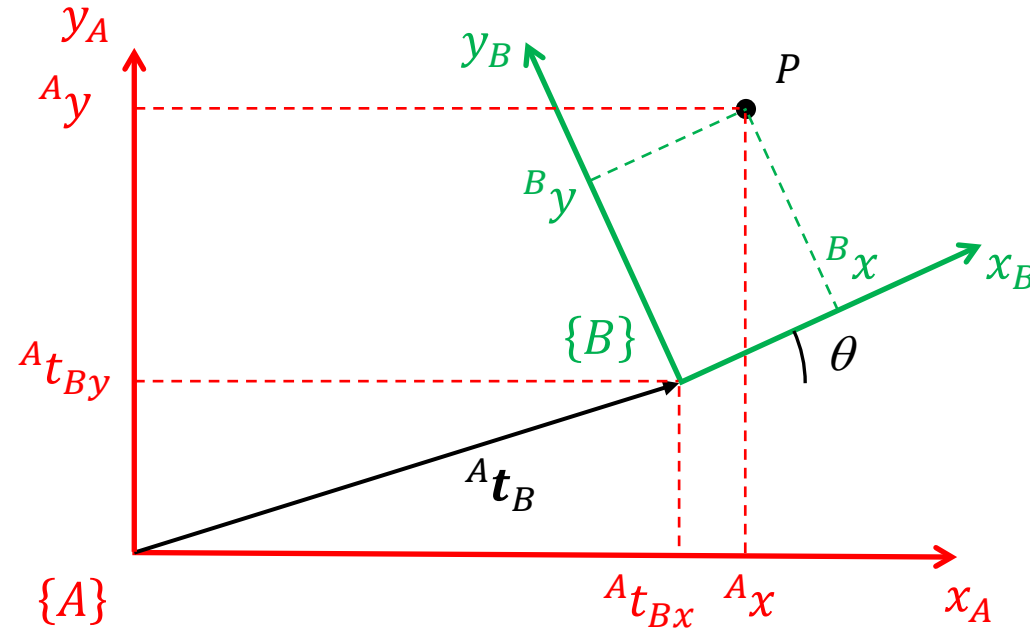
- Recall that we can think of ${}^A\xi_B$ as the translation and rotation required in order to make $\{A\}$ coincide with $\{B\}$

Investigating pose in 2D



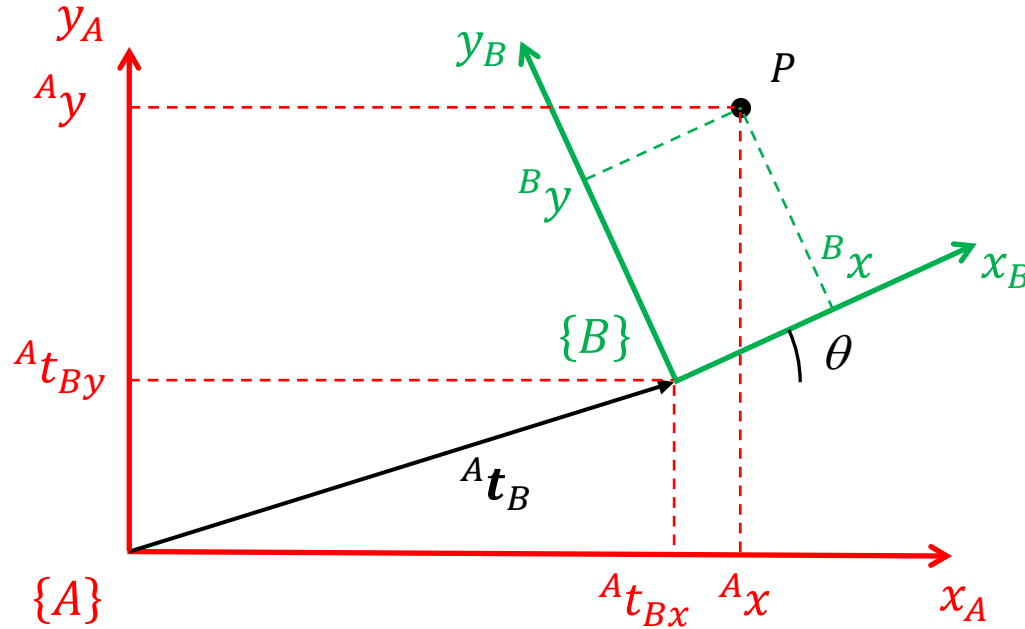
- Recall that we can think of ${}^A \xi_B$ as the translation and rotation required in order to make $\{A\}$ coincide with $\{B\}$
- To coincide with $\{B\}$, $\{A\}$ must undergo a translation ${}^A t_B$ and a rotation by an angle θ

Investigating pose in 2D



- Let $^A \mathbf{p} = [^A x, ^A y]^T$, $^B \mathbf{p} = [^B x, ^B y]^T$ and $^A \mathbf{t}_B = [^A t_{Bx}, ^A t_{By}]^T$

Investigating pose in 2D

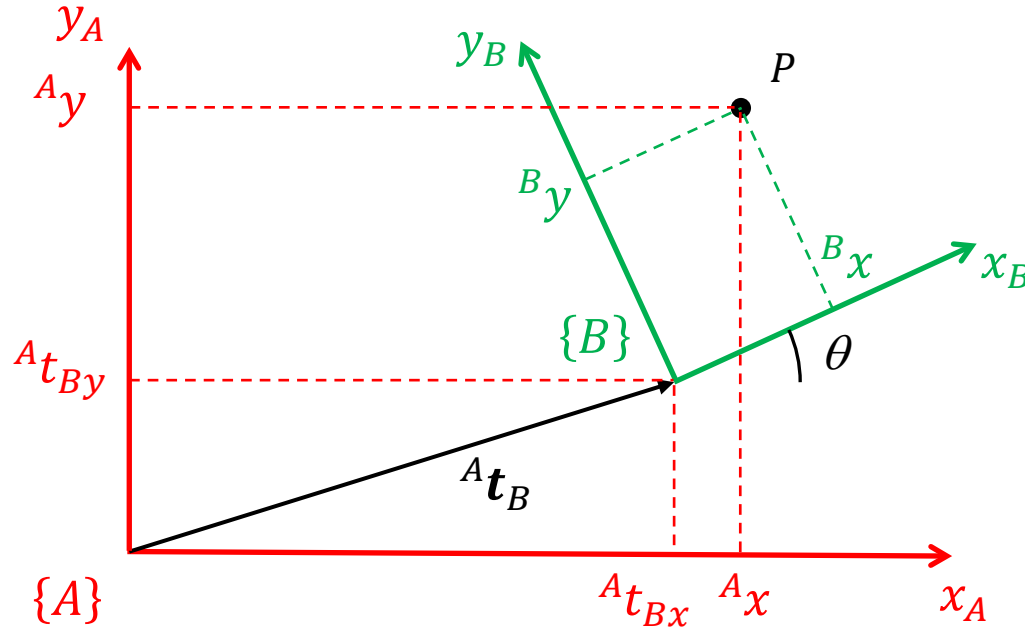


- Let ${}^A\mathbf{p} = [{}^Ax, {}^Ay]^T$, ${}^B\mathbf{p} = [{}^Bx, {}^By]^T$ and ${}^A\mathbf{t}_B = [{}^At_{Bx}, {}^At_{By}]^T$
- From the figure we can see that

$${}^Ax = {}^At_{Bx} + {}^Bx \cos \theta - {}^By \sin \theta$$

$${}^Ay = {}^At_{By} + {}^Bx \sin \theta + {}^By \cos \theta$$

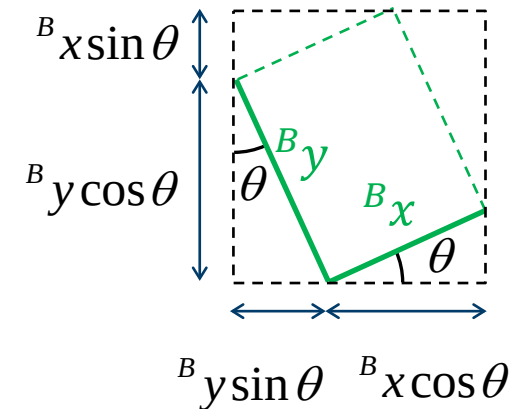
Investigating pose in 2D



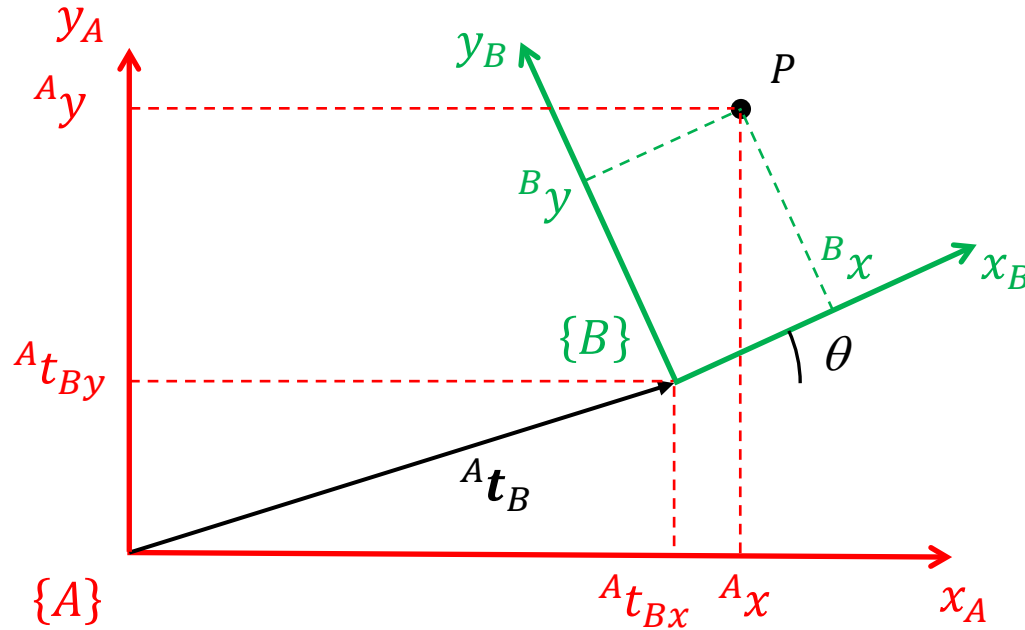
- Let ${}^A\mathbf{p} = [{}^A x, {}^A y]^T$, ${}^B\mathbf{p} = [{}^B x, {}^B y]^T$ and ${}^A\mathbf{t}_B = [{}^A t_{Bx}, {}^A t_{By}]^T$
- From the figure we can see that

$${}^A x = {}^A t_{Bx} + {}^B x \cos \theta - {}^B y \sin \theta$$

$${}^A y = {}^A t_{By} + {}^B x \sin \theta + {}^B y \cos \theta$$



Investigating pose in 2D



- In matrix form

$$\begin{bmatrix} {}^A x \\ {}^A y \end{bmatrix} = \begin{bmatrix} {}^A t_{Bx} \\ {}^A t_{By} \end{bmatrix} + \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} {}^B x \\ {}^B y \end{bmatrix}$$

$${}^A \mathbf{p} = {}^A \mathbf{t}_B + {}^A R_B {}^B \mathbf{p}$$

Investigating pose in 2D

$$\begin{bmatrix} {}^A x \\ {}^A y \end{bmatrix} = \begin{bmatrix} {}^A t_{Bx} \\ {}^A t_{By} \end{bmatrix} + \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} {}^B x \\ {}^B y \end{bmatrix}$$
$${}^A \mathbf{p} = {}^A \mathbf{t}_B + {}^A R_B {}^B \mathbf{p}$$

- Can we represent the pose ${}^A \xi_B$ by the pair $({}^A R_B, {}^A \mathbf{t}_B)$?

Investigating pose in 2D

$$\begin{bmatrix} {}^A x \\ {}^A y \end{bmatrix} = \begin{bmatrix} {}^A t_{Bx} \\ {}^A t_{By} \end{bmatrix} + \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} {}^B x \\ {}^B y \end{bmatrix}$$
$${}^A \mathbf{p} = {}^A \mathbf{t}_B + {}^A R_B {}^B \mathbf{p}$$

- Can we represent the pose ${}^A \xi_B$ by the pair $({}^A R_B, {}^A \mathbf{t}_B)$?

$$\begin{aligned} {}^A \mathbf{p} &= {}^A \xi_B \cdot {}^B \mathbf{p} &\mapsto & {}^A \mathbf{p} = {}^A R_B {}^B \mathbf{p} + {}^A \mathbf{t}_B \\ {}^A \xi_C &= {}^A \xi_B \oplus {}^B \xi_C &\mapsto & ({}^A R_C, {}^A \mathbf{t}_C) = ({}^A R_B {}^B R_C, {}^A R_B {}^B \mathbf{t}_C + {}^A \mathbf{t}_B) \\ \ominus {}^A \xi_B &&\mapsto & ({}^A R_C^T, -{}^A R_C^T {}^A \mathbf{t}_C) \end{aligned}$$

Investigating pose in 2D

$$\begin{bmatrix} {}^A x \\ {}^A y \end{bmatrix} = \begin{bmatrix} {}^A t_{Bx} \\ {}^A t_{By} \end{bmatrix} + \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} {}^B x \\ {}^B y \end{bmatrix}$$
$${}^A \mathbf{p} = {}^A \mathbf{t}_B + {}^A R_B {}^B \mathbf{p}$$

- Can we represent the pose ${}^A \xi_B$ by the pair $({}^A R_B, {}^A \mathbf{t}_B)$?

$$\begin{aligned} {}^A \mathbf{p} &= {}^A \xi_B \cdot {}^B \mathbf{p} &\mapsto & {}^A \mathbf{p} = {}^A R_B {}^B \mathbf{p} + {}^A \mathbf{t}_B \\ {}^A \xi_C &= {}^A \xi_B \oplus {}^B \xi_C &\mapsto & ({}^A R_C, {}^A \mathbf{t}_C) = ({}^A R_B {}^B R_C, {}^A R_B {}^B \mathbf{t}_C + {}^A \mathbf{t}_B) \\ \ominus {}^A \xi_B &&\mapsto & ({}^A R_C^T, -{}^A R_C^T {}^A \mathbf{t}_C) \end{aligned}$$

- Yes, but there is a better option!

Investigating pose in 2D

- Observe the following equivalence

$$\begin{aligned} \begin{bmatrix} {}^A x \\ {}^A y \end{bmatrix} &= {}^A R_B \begin{bmatrix} {}^B x \\ {}^B y \end{bmatrix} + {}^A \mathbf{t}_B & \Leftrightarrow & \begin{bmatrix} {}^A x \\ {}^A y \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R_B & {}^A \mathbf{t}_B \\ \mathbf{0}_{1 \times 2} & 1 \end{bmatrix} \begin{bmatrix} {}^B x \\ {}^B y \\ 1 \end{bmatrix} \\ {}^A \mathbf{p} &= {}^A R_B {}^B \mathbf{p} + {}^A \mathbf{t}_B & \Leftrightarrow & {}^A \tilde{\mathbf{p}} = {}^A T_B {}^B \tilde{\mathbf{p}} \end{aligned}$$

Investigating pose in 2D

- Observe the following equivalence

$$\begin{aligned} \begin{bmatrix} {}^A x \\ {}^A y \end{bmatrix} &= {}^A R_B \begin{bmatrix} {}^B x \\ {}^B y \end{bmatrix} + {}^A \mathbf{t}_B & \Leftrightarrow & \begin{bmatrix} {}^A x \\ {}^A y \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R_B & {}^A \mathbf{t}_B \\ \mathbf{0}_{1 \times 2} & 1 \end{bmatrix} \begin{bmatrix} {}^B x \\ {}^B y \\ 1 \end{bmatrix} \\ {}^A \mathbf{p} &= {}^A R_B {}^B \mathbf{p} + {}^A \mathbf{t}_B & \Leftrightarrow & {}^A \tilde{\mathbf{p}} = {}^A T_B {}^B \tilde{\mathbf{p}} \end{aligned}$$

- Can we represent the pose ${}^A \xi_B$ by the matrix ${}^A T_B$?

Investigating pose in 2D

- Observe the following equivalence

$$\begin{aligned} \begin{bmatrix} {}^A x \\ {}^A y \end{bmatrix} &= {}^A R_B \begin{bmatrix} {}^B x \\ {}^B y \end{bmatrix} + {}^A \mathbf{t}_B &\Leftrightarrow \begin{bmatrix} {}^A x \\ {}^A y \\ 1 \end{bmatrix} &= \begin{bmatrix} {}^A R_B & {}^A \mathbf{t}_B \\ \mathbf{0}_{1 \times 2} & 1 \end{bmatrix} \begin{bmatrix} {}^B x \\ {}^B y \\ 1 \end{bmatrix} \\ {}^A \mathbf{p} &= {}^A R_B {}^B \mathbf{p} + {}^A \mathbf{t}_B &\Leftrightarrow {}^A \tilde{\mathbf{p}} &= {}^A T_B {}^B \tilde{\mathbf{p}} \end{aligned}$$

- Can we represent the pose ${}^A \xi_B$ by the matrix ${}^A T_B$?

$$\begin{aligned} {}^A \mathbf{p} &= {}^A \xi_B \cdot {}^B \mathbf{p} &\mapsto & {}^A \tilde{\mathbf{p}} = {}^A T_B {}^B \tilde{\mathbf{p}} \\ {}^A \xi_C &= {}^A \xi_B \oplus {}^B \xi_C &\mapsto & {}^A T_C = {}^A T_B {}^B T_C \\ \ominus {}^A \xi_B &&\mapsto & {}^A T_B^{-1} \end{aligned}$$

- Yes, and the algebraic properties are nice!

Investigating pose in 2D

- But...

$$\underbrace{\begin{bmatrix} {}^A x \\ {}^A y \end{bmatrix}}_{{}^A \mathbf{p}} = {}^A R_B \underbrace{\begin{bmatrix} {}^B x \\ {}^B y \end{bmatrix}}_{{}^B \mathbf{p}} + {}^A \mathbf{t}_B \Leftrightarrow \underbrace{\begin{bmatrix} {}^A x \\ {}^A y \\ 1 \end{bmatrix}}_{{}^A \tilde{\mathbf{p}}} = \underbrace{\begin{bmatrix} {}^A R_B & {}^A \mathbf{t}_B \\ \mathbf{0}_{1 \times 2} & 1 \end{bmatrix}}_{{}^A T_B} \underbrace{\begin{bmatrix} {}^B x \\ {}^B y \\ 1 \end{bmatrix}}_{{}^B \tilde{\mathbf{p}}}$$

- We are describing points in the plane with 3 coordinates despite that they only have 2 degrees of freedom...
- The non linear transformation ${}^B \mathbf{p} \mapsto {}^A \mathbf{p}$ then becomes a linear transformation ${}^B \tilde{\mathbf{p}} \mapsto {}^A \tilde{\mathbf{p}}$
- What is going on?

Investigating pose in 2D

- We have “discovered” some basic constructions from projective geometry

$$\underbrace{\begin{bmatrix} {}^A x \\ {}^A y \end{bmatrix}}_{{}^A p} = {}^A R_B \underbrace{\begin{bmatrix} {}^B x \\ {}^B y \end{bmatrix}}_{{}^B p} + {}^A t_B$$
$$\Updownarrow$$
$$\underbrace{\begin{bmatrix} {}^A x \\ {}^A y \\ 1 \end{bmatrix}}_{{}^A \tilde{p}} = \underbrace{\begin{bmatrix} {}^A R_B & {}^A t_B \\ \mathbf{0}_{1 \times 2} & 1 \end{bmatrix}}_{{}^A T_B} \underbrace{\begin{bmatrix} {}^B x \\ {}^B y \\ 1 \end{bmatrix}}_{{}^B \tilde{p}}$$

Investigating pose in 2D

- We have “discovered” some basic constructions from projective geometry

Non-linear transformation
of the Euclidean plane $\mathbb{R}^2$

Points in the Euclidean
plane are described by
Cartesian coordinates

$$\underbrace{\begin{bmatrix} {}^A x \\ {}^A y \end{bmatrix}}_{{}^A p} = {}^A R_B \underbrace{\begin{bmatrix} {}^B x \\ {}^B y \end{bmatrix}}_{{}^B p} + {}^A t_B$$

$\Leftrightarrow$

$$\underbrace{\begin{bmatrix} {}^A x \\ {}^A y \\ 1 \end{bmatrix}}_{{}^A \tilde{p}} = \underbrace{\begin{bmatrix} {}^A R_B & {}^A t_B \\ \mathbf{0}_{1 \times 2} & 1 \end{bmatrix}}_{{}^A T_B} \underbrace{\begin{bmatrix} {}^B x \\ {}^B y \\ 1 \end{bmatrix}}_{{}^B \tilde{p}}$$

Investigating pose in 2D

- We have “discovered” some basic constructions from projective geometry

Non-linear transformation
of the Euclidean plane $\mathbb{R}^2$

Points in the Euclidean plane
are described by
Cartesian coordinates

$$\underbrace{\begin{bmatrix} {}^A x \\ {}^A y \end{bmatrix}}_{{}^A p} = {}^A R_B \underbrace{\begin{bmatrix} {}^B x \\ {}^B y \end{bmatrix}}_{{}^B p} + {}^A t_B$$

$\Updownarrow$

$$\underbrace{\begin{bmatrix} {}^A x \\ {}^A y \\ 1 \end{bmatrix}}_{{}^A \tilde{p}} = \underbrace{\begin{bmatrix} {}^A R_B & {}^A t_B \\ \mathbf{0}_{1 \times 2} & 1 \end{bmatrix}}_{{}^A T_B} \underbrace{\begin{bmatrix} {}^B x \\ {}^B y \\ 1 \end{bmatrix}}_{{}^B \tilde{p}}$$

Linear transformation
of the projective plane $\mathbb{P}^2$

Points in the projective plane
are described by homogeneous
coordinates

This means that they are only
unique up to scale, i.e.
 $(x, y, 1) = (\lambda x, \lambda y, \lambda) \forall \lambda \in \mathbb{R} \setminus \{0\}$

The matrix representing the
projective transformation is also
homogeneous, i.e.

$${}^A T_B = \lambda {}^A T_B \forall \lambda \in \mathbb{R} \setminus \{0\}_{37}$$

Investigating pose in 2D

- Euclidean geometry

- ${}^A\xi_B \mapsto ({}^A R_B, {}^A \mathbf{t}_B)$
- Complicated algebra

$$\begin{aligned} {}^A \mathbf{p} &= {}^A \xi_B \cdot {}^B \mathbf{p} &\mapsto & {}^A \mathbf{p} = {}^A R_B {}^B \mathbf{p} + {}^A \mathbf{t}_B \\ {}^A \xi_C &= {}^A \xi_B \oplus {}^B \xi_C &\mapsto & ({}^A R_C, {}^A \mathbf{t}_C) = ({}^A R_B {}^B R_C, {}^A R_B {}^B \mathbf{t}_C + {}^A \mathbf{t}_B) \\ \ominus {}^A \xi_B &&\mapsto & ({}^A R_C^T, -{}^A R_C^T {}^A \mathbf{t}_C) \end{aligned}$$

- Projective geometry

- ${}^A \xi_B \mapsto {}^A T_B = \begin{bmatrix} {}^A R_B & {}^A \mathbf{t}_B \\ \mathbf{0} & 1 \end{bmatrix}$
- Simple algebra

$$\begin{aligned} {}^A \mathbf{p} &= {}^A \xi_B \cdot {}^B \mathbf{p} &\mapsto & {}^A \tilde{\mathbf{p}} = {}^A T_B {}^B \tilde{\mathbf{p}} \\ {}^A \xi_C &= {}^A \xi_B \oplus {}^B \xi_C &\mapsto & {}^A T_C = {}^A T_B {}^B T_C \\ \ominus {}^A \xi_B &&\mapsto & {}^A T_B^{-1} \end{aligned}$$

- Many problems in computer vision are easier to express and solve if we choose to think of points and transformations in terms of projective geometry
 - Algebra and computations become simpler

Representing pose in 2D

- The pose of $\{B\}$ relative to $\{A\}$ can be represented by a homogeneous transformation ${}^A T_B \in SE(2)$

$${}^A \xi_B \mapsto {}^A T_B = \begin{bmatrix} {}^A R_B & {}^A \mathbf{t}_B \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & {}^A t_{Bx} \\ \sin \theta & \cos \theta & {}^A t_{By} \\ 0 & 0 & 1 \end{bmatrix}$$

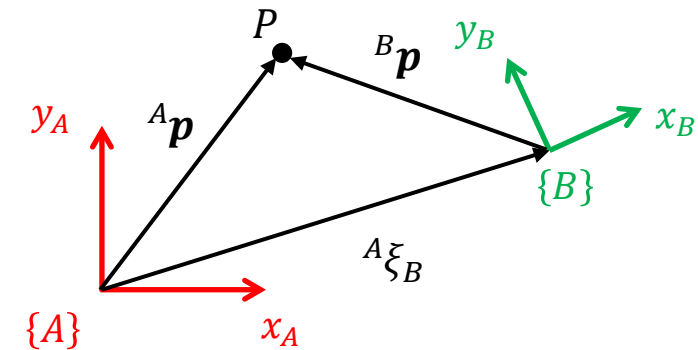
- Properties

$$\begin{aligned} {}^A \mathbf{p} &= {}^A \xi_B \cdot {}^B \mathbf{p} \mapsto {}^A \tilde{\mathbf{p}} = {}^A T_B {}^B \tilde{\mathbf{p}} \\ {}^A \xi_C &= {}^A \xi_B \oplus {}^B \xi_C \mapsto {}^A T_C = {}^A T_B {}^B T_C \\ \ominus {}^A \xi_B &\mapsto {}^A T_B^{-1} \end{aligned}$$

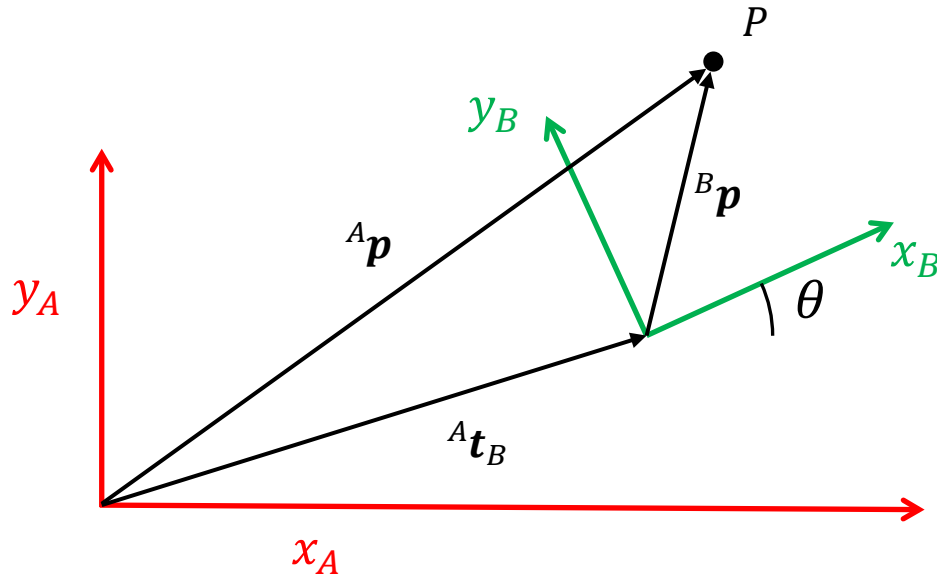
- Points are represented in homogeneous coordinates

$${}^A \tilde{\mathbf{p}} = {}^A T_B {}^B \tilde{\mathbf{p}}$$

$$\begin{bmatrix} {}^A x \\ {}^A y \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R_B & {}^A \mathbf{t}_B \\ \mathbf{0}_{1 \times 2} & 1 \end{bmatrix} \begin{bmatrix} {}^B x \\ {}^B y \\ 1 \end{bmatrix}$$



Example



- Let ${}^A\mathbf{t}_B = [4,1]^T$, ${}^B\mathbf{p} = [2,3]^T$ and $\theta = 27^\circ$
- Determine the pose of $\{B\}$ relative to $\{A\}$, i.e. AT_B
- Determine the coordinates of P in $\{A\}$, i.e. ${}^A\mathbf{p}$

Example

- From the previous slides we know that

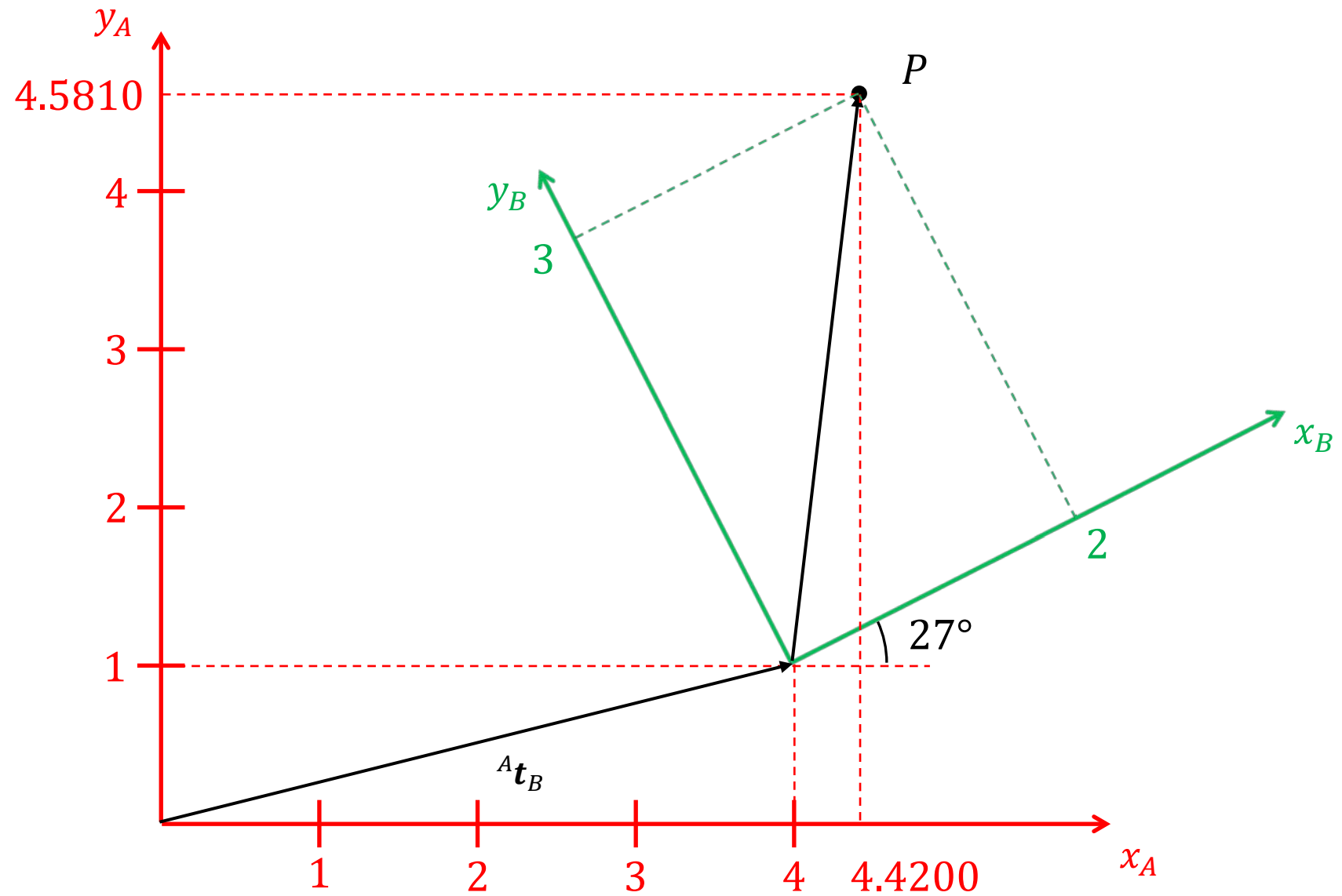
$${}^A T_B = \begin{bmatrix} \cos \theta & -\sin \theta & {}^A t_{Bx} \\ \sin \theta & \cos \theta & {}^A t_{By} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos 27^\circ & -\sin 27^\circ & 4 \\ \sin 27^\circ & \cos 27^\circ & 1 \\ 0 & 0 & 1 \end{bmatrix} \approx \begin{bmatrix} 0.8910 & -0.4540 & 4 \\ 0.4540 & 0.8910 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

- This allows us to compute ${}^A \tilde{\mathbf{p}}$

$${}^A \tilde{\mathbf{p}} = {}^A T_B {}^B \tilde{\mathbf{p}}$$
$$\begin{bmatrix} {}^A x \\ {}^A y \\ 1 \end{bmatrix} = \begin{bmatrix} 0.8910 & -0.4540 & 4 \\ 0.4540 & 0.8910 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4.4200 \\ 4.5810 \\ 1 \end{bmatrix} \Rightarrow {}^A \mathbf{p} = [4.4200 \quad 4.5810]^T$$

- This can also be verified by drawing

Example



Representing pose in 3D

- The pose of $\{B\}$ relative to $\{A\}$ can be represented by a homogeneous transformation ${}^A T_B \in SE(3)$

$${}^A \xi_B \mapsto {}^A T_B = \begin{bmatrix} {}^A R_B & {}^A \mathbf{t}_B \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & {}^A t_{Bx} \\ r_{21} & r_{22} & r_{23} & {}^A t_{By} \\ r_{31} & r_{32} & r_{33} & {}^A t_{Bz} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

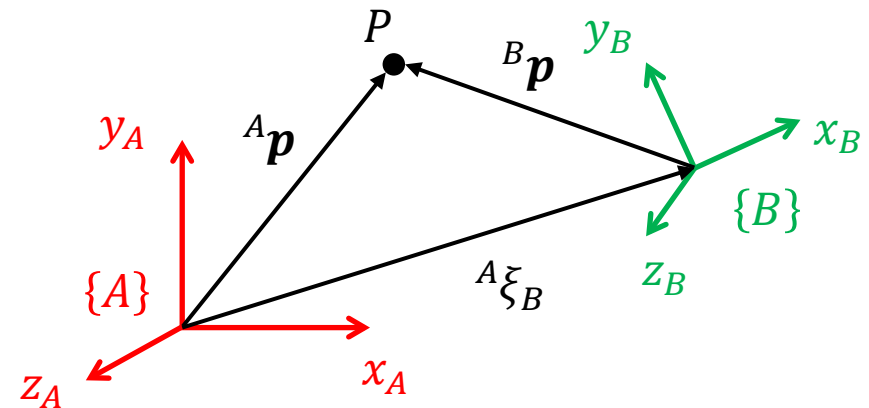
- Properties

$$\begin{aligned} {}^A \mathbf{p} = {}^A \xi_B \cdot {}^B \mathbf{p} &\mapsto {}^A \tilde{\mathbf{p}} = {}^A T_B {}^B \tilde{\mathbf{p}} \\ {}^A \xi_C = {}^A \xi_B \oplus {}^B \xi_C &\mapsto {}^A T_C = {}^A T_B {}^B T_C \\ \ominus {}^A \xi_B &\mapsto {}^A T_B^{-1} \end{aligned}$$

- Points are represented in homogeneous coordinates

$${}^A \tilde{\mathbf{p}} = {}^A T_B {}^B \tilde{\mathbf{p}}$$

$$\begin{bmatrix} {}^A x \\ {}^A y \\ {}^A z \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R_B & {}^A \mathbf{t}_B \\ \mathbf{0}_{1 \times 2} & 1 \end{bmatrix} \begin{bmatrix} {}^B x \\ {}^B y \\ {}^B z \\ 1 \end{bmatrix}$$



Representing pose in 3D

- The main difference between 3D and 2D is that rotation is far less intuitive in 3D
Also there are several different representations of rotation in 3D
 - Orthonormal rotation matrix $R \in SO(3)$
 - Euler angles $(\theta_1, \theta_2, \theta_3)$
 - Angle-axis $(\theta, \mathbf{e})$ or just $\boldsymbol{\theta} = \theta \mathbf{e}$
 - Unit quaternions $q = r + xi + yj + zk$
- Hence there are several ways to represent pose
 - Rotation matrix and translation vector $({}^A R_B, {}^A \mathbf{t}_B)$
 - Homogeneous transformation ${}^A T_B$
 - Euler angles and translation vector $(\theta_1, \theta_2, \theta_3, {}^A \mathbf{t}_B)$
 - Angle-axis and translation vector $(\theta, \mathbf{e}, {}^A \mathbf{t}_B)$
 - Unit quaternion and translation vector $({}^A q_B, {}^A \mathbf{t}_B)$

Representing rotation in 3D

- Orientation of $\{B\}$ relative to $\{A\}$
 - How $\{B\}$ should rotate to coincide with $\{A\}$

- Orthonormal rotation matrix ${}^A R_B \in SO(3)$

$${}^A R_B = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

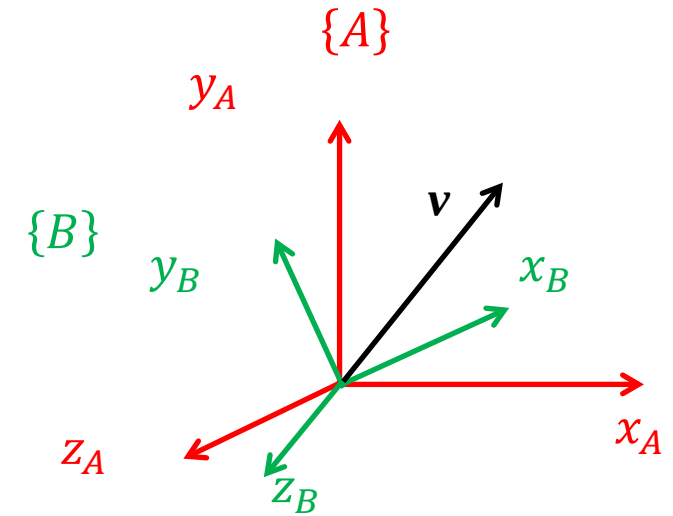
- Properties

$$\det({}^A R_B) = 1$$

$${}^A R_B^{-1} = {}^A R_B^T$$

$${}^A R_C = {}^A R_B {}^B R_C$$

$${}^A \mathbf{v} = {}^A R_B {}^B \mathbf{v}$$



- Nice algebraic properties
- Computations are simple
- Provides little intuitive understanding

Representing rotation in 3D

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Euler angles $(\theta_1, \theta_2, \theta_3)$ describes rotation in terms of basic rotations about 3 coordinate axis that must be specified
 - 12 compositions: $xyx, xzx, yxy, yzy, zxz, zyz, xyz, xzy, yzx, yxz, zxy, zyx$
 - E.g. $R_{xyz} = R_x(\theta_1)R_y(\theta_2)R_z(\theta_3)$

- Suffers from singularities
 - $\theta_2 = \pm \frac{\pi}{2}$ for sequences without repetition
 - $\theta_2 \in \{0, \pi\}$ for sequences with repetition
- It is possible to determine the Euler angles that corresponds to a given rotation matrix

$$\theta_y = \text{atan2}\left(r_{13}, \sqrt{r_{11}^2 + r_{12}^2}\right) \quad -\frac{\pi}{2} < \theta_y < \frac{\pi}{2}$$

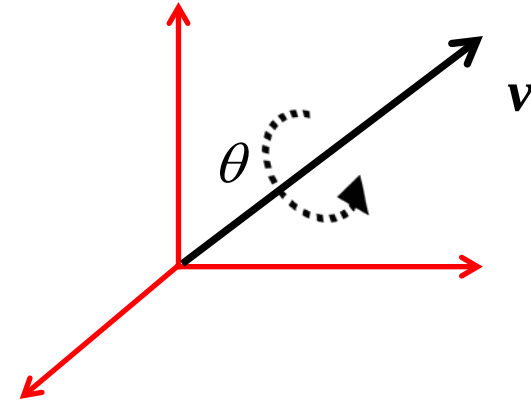
$$R_{xyz} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \Rightarrow \theta_x = \text{atan2}\left(-\frac{r_{23}}{\cos \theta_y}, \frac{r_{33}}{\cos \theta_y}\right) \quad -\pi < \theta_x < \pi$$

$$\theta_z = \text{atan2}\left(-\frac{r_{12}}{\cos \theta_y}, \frac{r_{11}}{\cos \theta_y}\right) \quad -\pi < \theta_z < \pi$$

Representing rotation in 3D

- An angle-axis pair $(\theta, \mathbf{v})$ describes rotation in terms of an angle and an axis of revolution
- Euler's rotation theorem states that any sequence of rotations in 3 dimensions is equivalent to a single rotation θ about an axis $\mathbf{v}$
- The rotation matrix R corresponding to an angle-axis pair $(\theta, \mathbf{v})$ can be found by Rodrigues' rotation formula

$$R = I_3 + \sin \theta [\mathbf{v}]_{\times} + (1 - \cos \theta)(\mathbf{v}\mathbf{v}^T - I_3)$$



Where $[\mathbf{v}]_{\times}$ is the matrix representation of the cross product

$$[\mathbf{v}]_{\times} = \begin{bmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{bmatrix}$$

- The angle-axis pair $(\theta, \mathbf{v})$ corresponding to a rotation matrix R can be found from the eigenvalues and eigenvectors of R

Representing rotation in 3D

- Unit quaternions are commonly used to work with rotation/orientation in 3D
- A quaternion $q \in \mathbb{H}$ is a number, with 1 real term and 3 imaginary terms

$$q = q_0 + q_1i + q_2j + q_3k = q_0 + \mathbf{v}$$

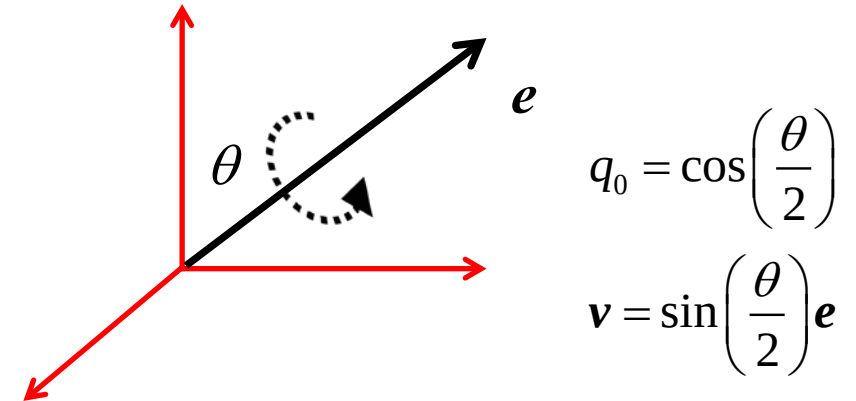
$$i^2 = j^2 = k^2 = ijk = -1$$

- Conjugate quaternion

$$q^* = q_0 - q_1i - q_2j - q_3k = q_0 - \mathbf{v}$$

- A unit quaternion satisfy

$$\|q\| = \sqrt{q^*q} = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2} = 1$$



- Quaternion multiplication

$$\begin{aligned} qp &= (q_0 + \mathbf{v})(p_0 + \mathbf{w}) \\ &= \underbrace{q_0p_0 - \mathbf{v} \cdot \mathbf{w}}_{\in \mathbb{R}} + \underbrace{q_0\mathbf{w} + p_0\mathbf{v} + \mathbf{v} \times \mathbf{w}}_{\in \mathbb{R}^3} \end{aligned}$$

- Composing two unit quaternions takes 16 multiplications and 12 additions, while composing two rotation matrixes takes 27 multiplications and 18 additions

Representing rotation in 3D

- The unit quaternion corresponding to the rotation matrix R is given by

$$q_0 = \frac{1}{2} \sqrt{1 + r_{11} + r_{22} + r_{33}}$$

$$q_1 = \frac{r_{32} - r_{23}}{4q_0}$$

$$q_2 = \frac{r_{13} - r_{31}}{4q_0}$$

$$q_3 = \frac{r_{21} - r_{12}}{4q_0}$$

- The rotation matrix corresponding to the unit quaternion $q = q_0 + q_1i + q_2j + q_3k$ is given by

$$R = \begin{bmatrix} 1 - 2q_2^2 - 2q_3^2 & 2(q_1q_2 - q_0q_3) & 2(q_1q_3 + q_0q_2) \\ 2(q_1q_2 + q_0q_3) & 1 - 2q_1^2 - 2q_3^2 & 2(q_2q_3 - q_0q_1) \\ 2(q_1q_3 - q_0q_2) & 2(q_2q_3 + q_0q_1) & 1 - 2q_1^2 - 2q_2^2 \end{bmatrix}$$

- SLERP – Spherical linear interpolation
An algorithm for determining a rotation partway between two given rotations

Useful for visualizing rotations

Summary

- Pose
 - General properties
- Representing pose in 2D
 - Homogeneous transformations
- Representing pose in 3D
 - Homogeneous transformations
 - Other alternatives
- Representing rotation in 3D
 - Rotation matrix
 - Euler angles
 - Angle-axis
 - Unit quaternion
- Additional information
 - Szeliski 2.1.1, 2.1.2, 2.1.4